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Okajima

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(54) **WORK MACHINE AND METHOD FOR CALCULATING DEGREE OF INCLINATION OF WORK MACHINE**

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E02F 9/20	(2006.01)
E02F 9/26	(2006.01)

(Continued)

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CPC ***E02F 9/24*** (2013.01); ***E02F 9/166*** (2013.01); ***E02F 9/2033*** (2013.01); ***E02F 9/26*** (2013.01); ***G01C 9/18*** (2013.01); *E02F 3/32* (2013.01)

(58) **Field of Classification Search**

CPC ... E02F 3/32; E02F 9/163; E02F 9/166; E02F 9/2033; E02F 9/24; E02F 9/26; E02F 9/264; G01C 9/18

See application file for complete search history.

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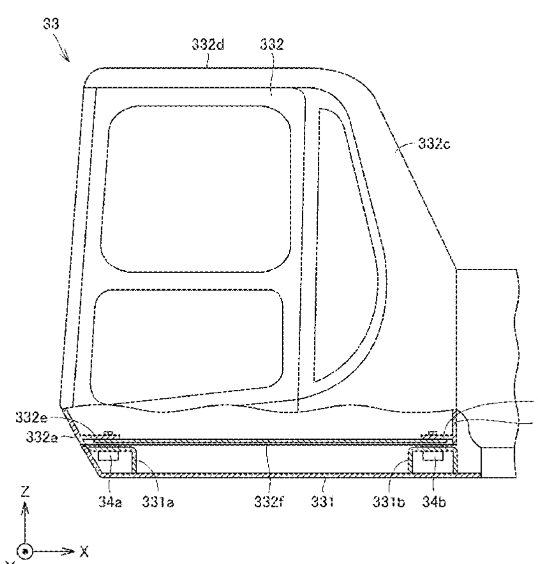
(57) **ABSTRACT**

A work machine includes a body, a work implement, and a controller. The body includes a frame to which the work implement is attached, an operator's cab, and a plurality of liquid-sealed mounts that are installed at different positions of the frame and support the operator's cab. At least one of the plurality of liquid-sealed mounts includes a sensor that detects a liquid pressure of a sealed liquid. The controller calculates an inclination degree of the work machine with respect to a horizontal plane based on a result of the detection by the sensor.

10 Claims, 9 Drawing Sheets



(51) **Int. Cl.**
G01C 9/18 (2006.01)
E02F 3/32 (2006.01)

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FIG.1

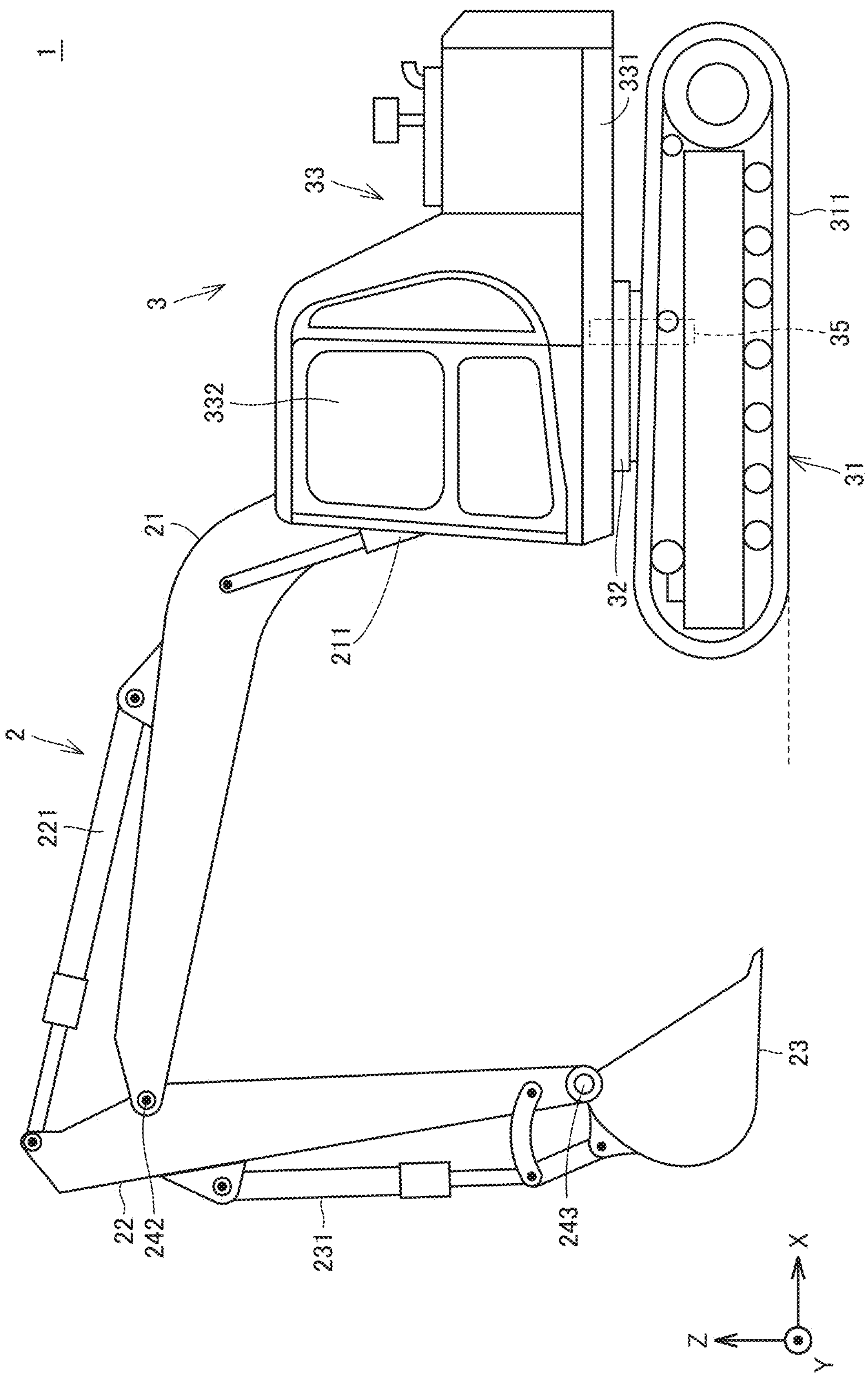


FIG.2

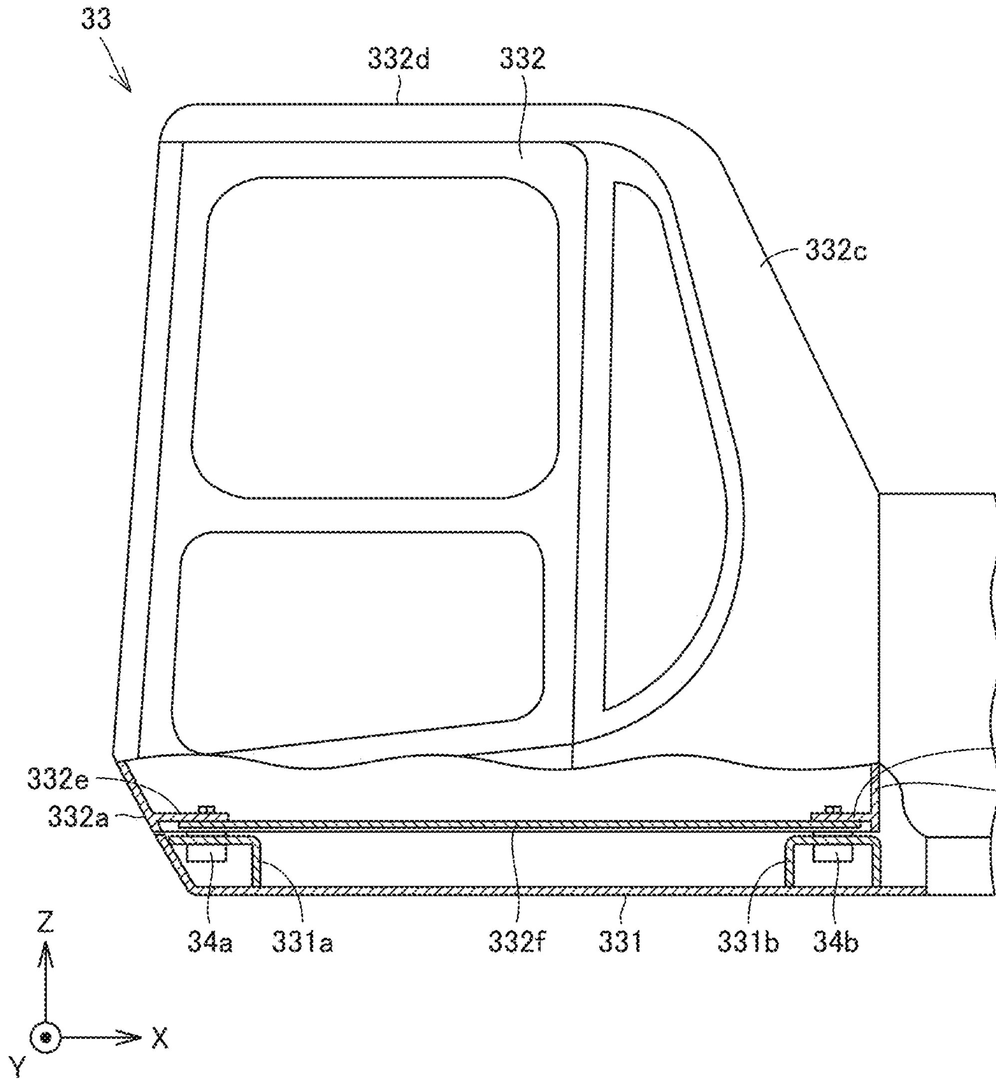

33
332d
332
332c
332e
332b
332e
332a
34a
331a
332f
331
331b
34b
Z
X
Y

FIG.3

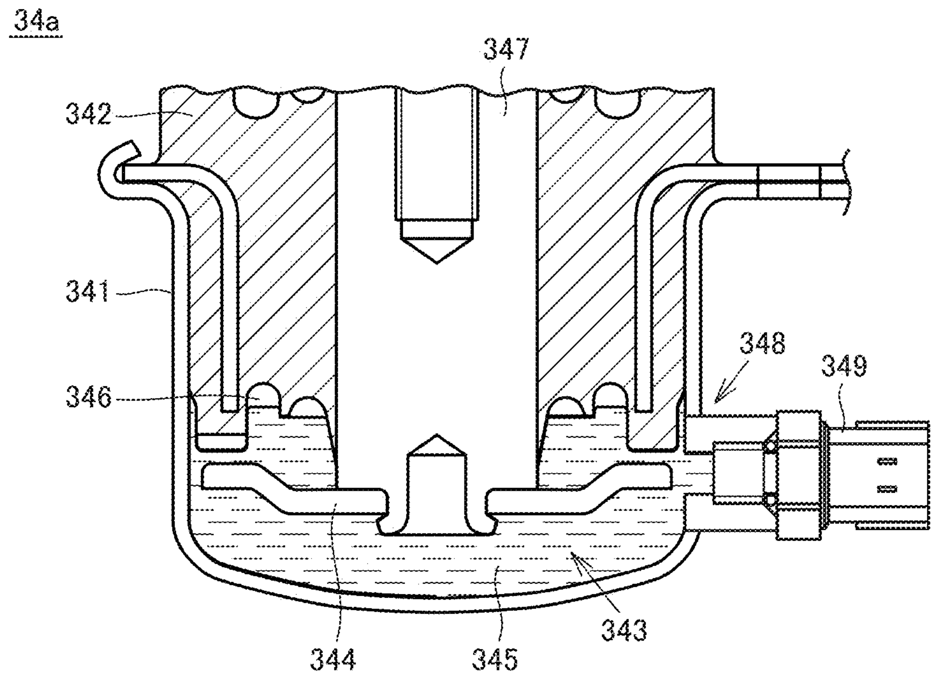


FIG.4

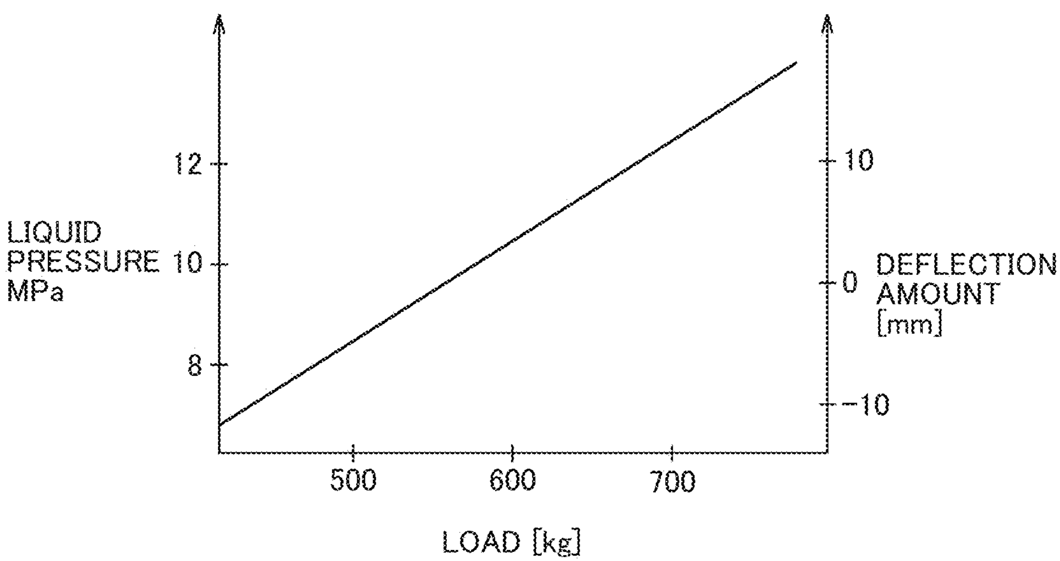

12
LIQUID
PRESSURE
MPa
10
8
10
0
DEFLECTION
AMOUNT
[mm]
−10
500
600
700
LOAD [kg]

FIG.5A

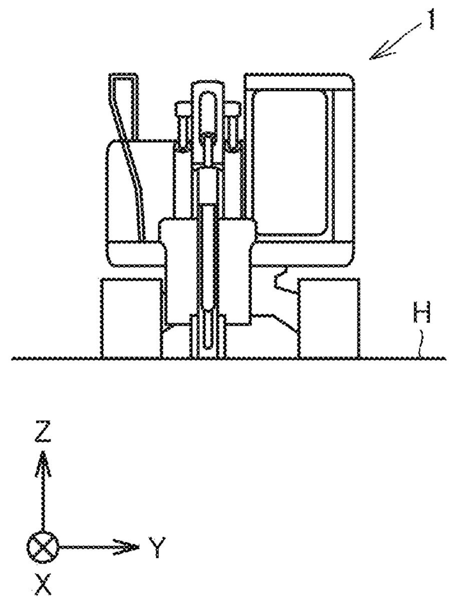


FIG.5B

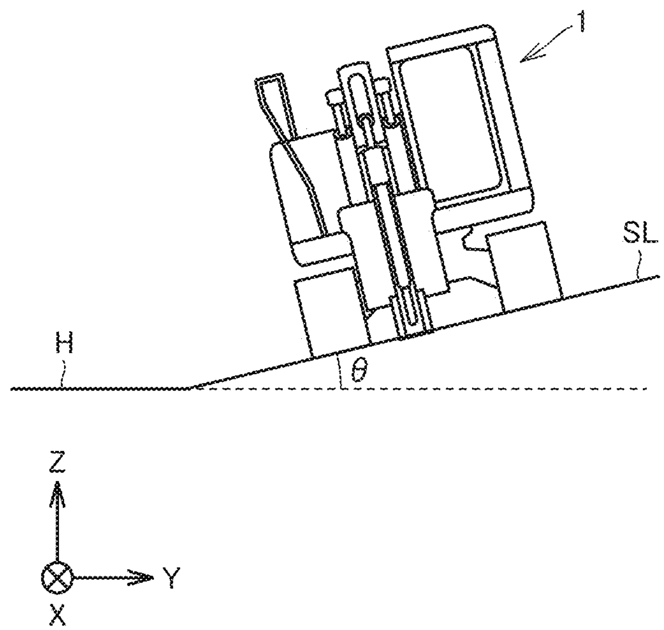


FIG.6

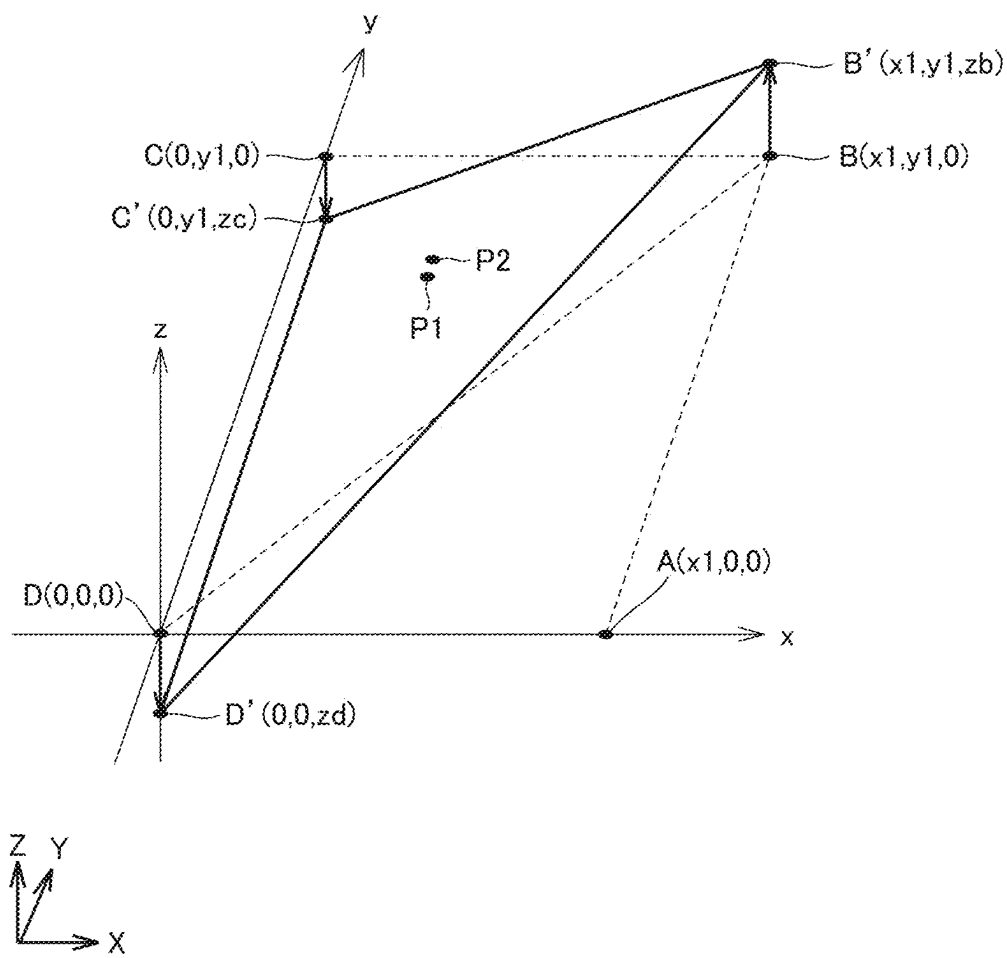


FIG.7

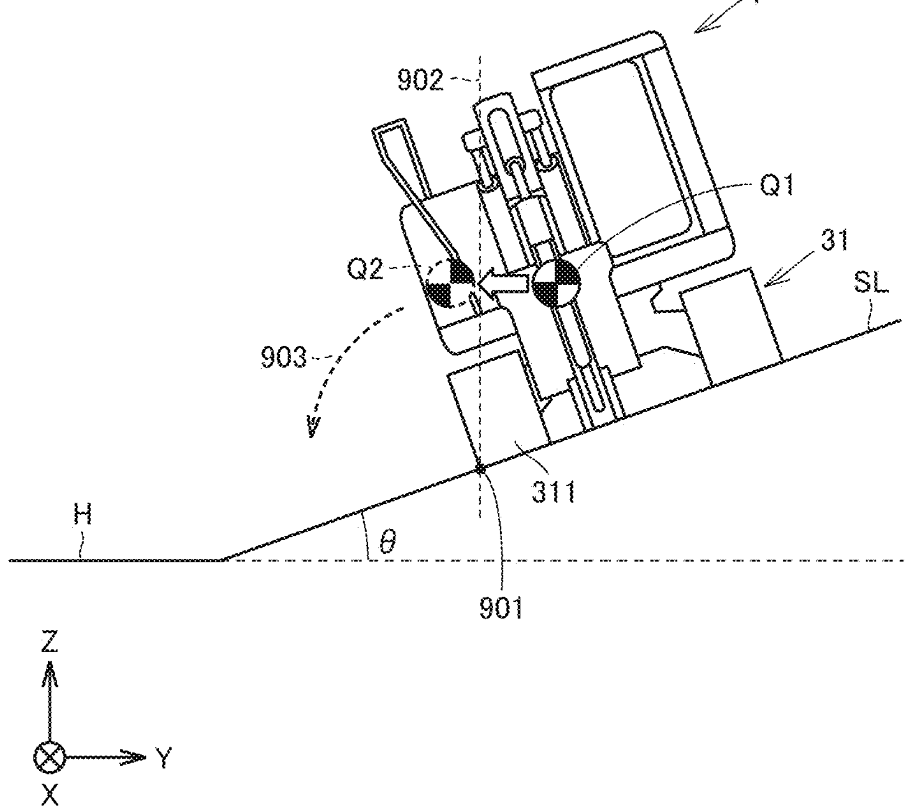


FIG.8

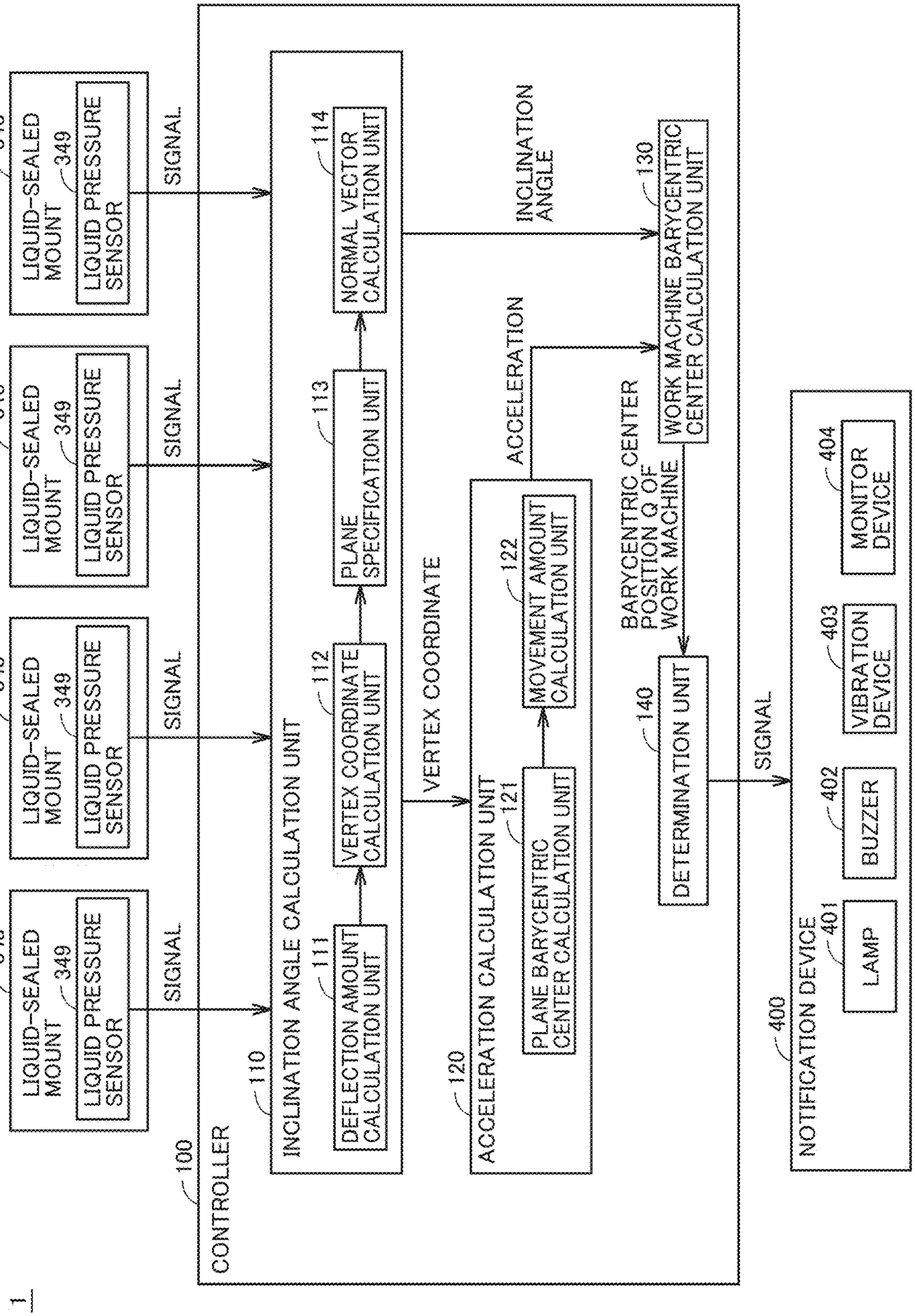
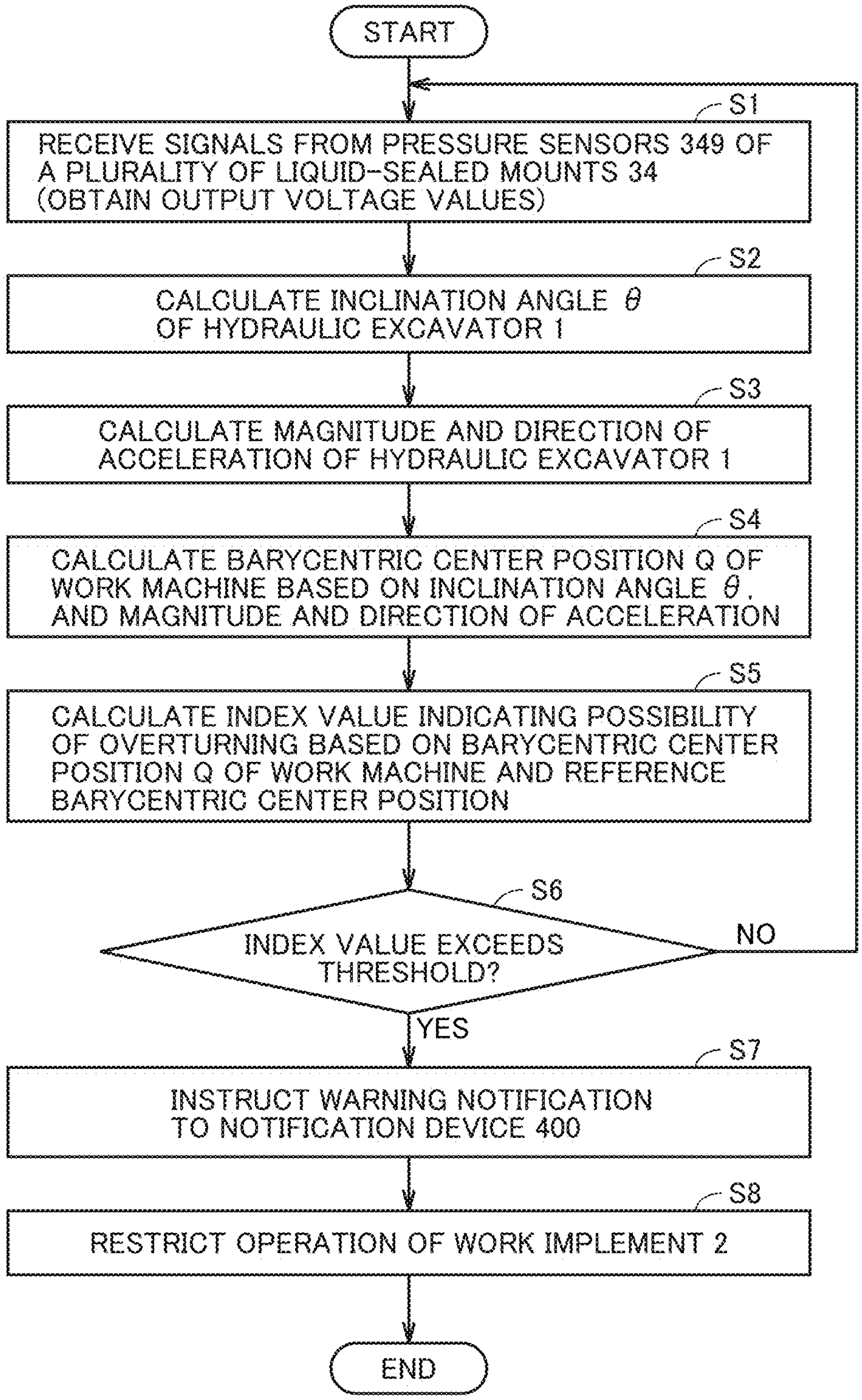

1
100
CONTROLLER
34a
LIQUID-SEALED MOUNT
349
LIQUID PRESSURE SENSOR
34b
LIQUID-SEALED MOUNT
349
LIQUID PRESSURE SENSOR
34c
LIQUID-SEALED MOUNT
349
LIQUID PRESSURE SENSOR
34d
LIQUID-SEALED MOUNT
349
LIQUID PRESSURE SENSOR
SIGNAL
SIGNAL
SIGNAL
SIGNAL
110
INCLINATION ANGLE CALCULATION UNIT
111
DEFLECTION AMOUNT CALCULATION UNIT
112
VERTEX COORDINATE CALCULATION UNIT
113
PLANE SPECIFICATION UNIT
114
NORMAL VECTOR CALCULATION UNIT
VERTEX COORDINATE
INCLINATION ANGLE
120
ACCELERATION CALCULATION UNIT
121
PLANE BARYCENTRIC CENTER CALCULATION UNIT
122
MOVEMENT AMOUNT CALCULATION UNIT
ACCELERATION
130
WORK MACHINE BARYCENTRIC CENTER CALCULATION UNIT
BARYCENTRIC CENTER POSITION Q OF WORK MACHINE
140
DETERMINATION UNIT
SIGNAL
400
NOTIFICATION DEVICE
401
LAMP
402
BUZZER
403
VIBRATION DEVICE
404
MONITOR DEVICE

FIG.9


START
S1
RECEIVE SIGNALS FROM PRESSURE SENSORS 349 OF A PLURALITY OF LIQUID-SEALED MOUNTS 34 (OBTAIN OUTPUT VOLTAGE VALUES)
S2
CALCULATE INCLINATION ANGLE θ OF HYDRAULIC EXCAVATOR 1
S3
CALCULATE MAGNITUDE AND DIRECTION OF ACCELERATION OF HYDRAULIC EXCAVATOR 1
S4
CALCULATE BARYCENTRIC CENTER POSITION Q OF WORK MACHINE BASED ON INCLINATION ANGLE θ, AND MAGNITUDE AND DIRECTION OF ACCELERATION
S5
CALCULATE INDEX VALUE INDICATING POSSIBILITY OF OVERTURNING BASED ON BARYCENTRIC CENTER POSITION Q OF WORK MACHINE AND REFERENCE BARYCENTRIC CENTER POSITION
S6
INDEX VALUE EXCEEDS THRESHOLD?
NO
YES
S7
INSTRUCT WARNING NOTIFICATION TO NOTIFICATION DEVICE 400
S8
RESTRICT OPERATION OF WORK IMPLEMENT 2
END

WORK MACHINE AND METHOD FOR CALCULATING DEGREE OF INCLINATION OF WORK MACHINE

TECHNICAL FIELD

The present disclosure relates to a work machine and a work machine inclination degree calculation method.

BACKGROUND ART

Conventionally, a work machine including a work implement is known. For example, Japanese Patent Laying-Open No. H07-207711 (PTL 1) discloses a hydraulic excavator including an overturning prevention device as the work machine. Each detection signal from boom angle detection means, dipper stick angle detection means, revolving angle detection means, and inclination angle detection means is input to a controller of the overturning prevention device. The controller generates an alarm, stops an operation of a boom and a dipper stick, and stops a turning operation before a risk of overturning is generated according to a revolving operation of the boom and the dipper stick and the revolving operation of an upper revolving unit. An inclination angle sensor that detects an inclination angle of a machine body with respect to a horizontal plane is used as the inclination angle detector.

CITATION LIST

Patent Literature

PTL 1: Japanese Patent Laying-Open No. H07-207711

SUMMARY OF INVENTION

Technical Problem

In PTL 1, the inclination angle sensor detects the inclination angle of the horizontal plane of the machine body of the work machine. Thus, the inclination angle sensor of PTL 1 is required to have high accuracy and durability.

The present disclosure provides a work machine and a work machine inclination degree calculation method capable of accurately determining the inclination degree of the work machine required to have durability with respect to the horizontal plane.

Solution to Problem

According to one aspect of the present disclosure, a work machine includes a body, a work implement, and a controller. The body includes a frame to which the work implement is attached, an operator's cab, and a plurality of liquid-sealed mounts that are installed at different positions of the frame and support the operator's cab. At least one of the plurality of liquid-sealed mounts includes a sensor that detects a liquid pressure of a sealed liquid. The controller calculates an inclination degree of the work machine with respect to a horizontal plane based on a result of the detection by the sensor.

According to another aspect of the present disclosure, a body of a work machine includes a frame to which a work implement is attached, an operator's cab, and a plurality of liquid-sealed mounts that are installed at different positions of the frame and support the operator's cab. A method for calculating an inclination degree of a work machine includes: detecting a liquid pressure of a liquid sealed in at least one of the plurality of liquid-sealed mounts; and calculating, by a controller of the work machine, an inclination degree of the work machine with respect to a horizontal plane based on a result of the detection.

Advantageous Effects of Invention

According to the present disclosure, the inclination degree of the work machine with respect to the horizontal plane can be accurately determined.

BRIEF DESCRIPTION OF DRAWINGS

FIG. **1** is a side view schematically illustrating a configuration of the hydraulic excavator.

FIG. **2** is a view illustrating a configuration of a revolving unit.

FIG. **3** is a view illustrating a main part of a liquid-sealed mount.

FIG. **4** is a view illustrating a characteristic of the liquid-sealed mount.

FIG. **5**A is a view illustrating a state in which the hydraulic excavator is positioned on a horizontal plane.

FIG. **5**B is a view illustrating a state in which the hydraulic excavator is positioned on an inclined plane.

FIG. **6** is a view illustrating changes in deflection amounts (displacement amounts) of three liquid-sealed mounts.

FIG. **7** is a view illustrating overturning of the hydraulic excavator.

FIG. **8** is a block diagram illustrating a hardware configuration of the hydraulic excavator and a functional configuration of a controller of the hydraulic excavator.

FIG. **9** is a flowchart illustrating a flow of processing of the controller.

DESCRIPTION OF EMBODIMENTS

Hereinafter, an embodiment will be described with reference to the drawings. It is planned from the beginning that structures in the embodiment are used while combined as appropriate. Some components may not be used.

In the following description, a hydraulic excavator will be described as an example of the work machine. However, the work machine is not limited to the hydraulic excavator, but is not particularly limited as long as it is the work machine having a work implement. In the following description, "upper", "lower", "front", "rear", "left" and "right" are terms based on an operator seated in a driver's seat.

FIG. **1** is a side view schematically illustrating a configuration of the hydraulic excavator.

As illustrated in FIG. **1**, a hydraulic excavator **1** includes a work implement **2** and a body **3**. Body **3** includes a traveling unit **31**, a swing circle **32**, a revolving unit **33**, and a hydraulic motor **35**.

Traveling unit **31** includes a pair of right and left crawler belt apparatuses **311**. Each of the pair of left and right crawler belt apparatuses **311** includes a crawler belt. When the pair of right and left crawler belts is rotationally driven, hydraulic excavator **1** self-propels.

Swing circle **32** is connected to hydraulic motor **35**. Swing circle **32** is rotated by rotational driving of hydraulic motor **35**.

Revolving unit **33** is installed on traveling unit **31** with swing circle **32** interposed therebetween. Revolving unit **33** turns with respect to traveling unit **31** as swing circle **32** rotates.

Revolving unit **33** includes a frame **331** to which work implement **2** is attached, an operator's cab **332**, and a controller (see FIG. **8**) that controls the operation of hydraulic excavator **1**. For example, operator's cab **332** is disposed on a front left side (vehicle front side) of revolving unit **33**. Hydraulic motor **35** is driven by hydraulic oil supplied from a hydraulic pressure source.

Work implement **2** is pivotally supported on the front side of revolving unit **33**, for example, on the right side of operator's cab **332**. Work implement **2** includes a boom **21**, a dipper stick **22**, a bucket **23**, a boom cylinder **211**, a dipper stick cylinder **221**, and a bucket cylinder **231**.

Boom **21** is attached to revolving unit **33**. A proximal end portion of boom **21** is rotatably connected to revolving unit **33** by a boom foot pin (not illustrated).

Dipper stick **22** is attached to the distal end of boom **21**. The proximal end portion of dipper stick **22** is rotatably connected to a distal end portion of boom **21** by a boom distal end pin **242**.

Bucket **23** is attached to the distal end of dipper stick **22**. Bucket **23** is rotatably connected to the distal end portion of dipper stick **22** by a dipper stick distal end pin **243**.

Boom **21** can be driven by boom cylinder **211**. Boom cylinder **211** is driven by the hydraulic oil supplied from a hydraulic pressure source (a hydraulic pump and an oil tank (not illustrated)). With this driving, boom **21** is vertically turnable about the boom foot pin (not illustrated) with respect to revolving unit **33**.

Dipper stick **22** can be driven by dipper stick cylinder **221**. By this driving, dipper stick **22** is vertically turnable with respect to boom **21** about boom distal end pin **242**.

Bucket **23** can be driven by bucket cylinder **231**. By this driving, bucket **23** is turnable in the vertical direction with respect to dipper stick **22** about dipper stick distal end pin **243**. In this manner, work implement **2** can be driven.

FIG. **2** is a view illustrating a configuration of revolving unit **33**.

As illustrated in FIG. **2**, revolving unit **33** includes frame **331** and operator's cab **332** as described above. Revolving unit **33** further includes a plurality of liquid-sealed mounts **34***a*, **34***b*, ... The liquid-sealed mount is also referred to as "viscus mount" or "anti-vibration mount".

Frame **331** includes a frame body **331***a* and a beam **331***b* extending in the left-right direction.

Operator's cab **332** includes a front surface portion **332***a*, a rear surface portion **332***b*, a side surface portion **332***c* (only the left side is illustrated), and an upper surface portion **332***d*. Operator's cab **332** is surrounded by portions **332***a*, **332***b*, **332***c*, **332***d*. Operator's cab **332** is provided with a floor board bracket **332***e* on the lower end side. A floor board **332***f* forming the bottom of operator's cab **332** is attached to floor board bracket **332***e*.

The plurality of liquid-sealed mounts **34***a*, **34***b*, . . . are installed at different positions of frame **331** and support operator's cab **332**. In this example, revolving unit **33** includes four liquid-sealed mounts **34***a*, **34***b*, **34***c*, **34***d* (FIGS. **6** and **8**).

Liquid-sealed mount **34***a* is installed on the left side of the front portion of frame **331**. Liquid-sealed mount **34***b* is installed on the left side of the rear portion of frame **331**. Similarly, liquid-sealed mounts **34***c*, **34***d* (FIG. **6**) are installed on the right side of the rear portion of frame **331** and the right side of the front portion of frame **331**, respectively.

Liquid-sealed mounts **34***a*, **34***d* are disposed apart from each other in the left and right directions between frame body **331***a* and floor board **332***f* of operator's cab **332**. Liquid-sealed mounts **34***b*, **34***c* are disposed apart from each other in the left and right directions between beam **331***b* and floor board **332***f*.

In this manner, four liquid-sealed mounts **34***a* to **34***d* are installed at four corners of the lower portion of operator's cab **332**. Liquid-sealed mounts **34***a* to **34***d* can prevent transmission of vibration of frame **331** to operator's cab **332**.

FIG. **3** is a view illustrating a main part of liquid-sealed mount **34***a*.

As illustrated in FIG. **3**, liquid-sealed mount **34***a* includes a casing **341**, an elastic body **342**, a fluid chamber **343**, a resistance plate **344**, a shaft **347**, and a liquid pressure sensor **349**.

Casing **341** has an opening on the upper end side. Casing **341** includes a mounting port **348** in a side portion. Typically, mounting port **348** has a threaded groove.

Elastic body **342** is fixed to casing **341** or the like while the opening of casing **341** is closed. Elastic body **342** is typically rubber.

Shaft **347** is fixed to elastic body **342**. Shaft **347** is formed in a columnar shape extending in the axial direction. Shaft **347** is disposed at the center of casing **341** by fixing an intermediate region in the axial direction to elastic body **342**.

Casing **341**, elastic body **342**, and shaft **347** form a fluid chamber **343**. A viscous liquid **345** and a compressed air **346** are sealed in fluid chamber **343**. Viscous liquid **345** is typically silicon oil.

Resistance plate **344** is always immersed in viscous liquid **345** sealed in fluid chamber **343**. Resistance plate **344** moves in viscous liquid **345** together with shaft **347** when elastic body **342** is elastically deformed. The movement generates resistance force (damping force) against the movement of shaft **347**.

Liquid pressure sensor **349** is attached to mounting port **348** of casing **341**. Typically, a thread groove is formed at an end of liquid pressure sensor **349**. When the end of liquid pressure sensor **349** is screwed into mounting port **348** of casing **341**, liquid pressure sensor **349** is fixed to casing **341**.

Liquid pressure sensor **349** detects a liquid pressure of viscous liquid **345**. Liquid pressure sensor **349** notifies controller **100** (FIG. **8**) that controls the operation of hydraulic excavator **1** of a detection result. Liquid pressure sensor **349** periodically detects the liquid pressure and periodically transmits a detection result to controller **100**. Liquid pressure sensor **349** notifies controller **100** of an electric signal that is the detection result. In this example, liquid pressure sensor **349** outputs a voltage value to controller **100**.

As described above, liquid-sealed mount **34***a* includes liquid pressure sensor **349** that detects the liquid pressure of the viscous liquid sealed in liquid-sealed mount **34***a* and outputs the liquid pressure to controller **100**.

Because liquid-sealed mounts **34***b*, **34***c*, **34***d* also have the same configuration as liquid-sealed mount **34***a*, liquid-sealed mounts **34***b* to **34***d* will not be described again here. Hereinafter, when liquid-sealed mounts **34***a* to **34***d* are not distinguished, any one of them is also referred to as "liquid-sealed mount **34**".

FIG. **4** is a view illustrating a characteristic of liquid-sealed mount **34**.

As illustrated in FIG. **4**, the liquid pressure of viscous liquid **345** and a deflection amount (deformation amount) of elastic body **342** are proportional to a load applied to liquid-sealed mount **34**. As the load increases, the liquid pressure of viscous liquid **345** increases. When the load increases, the deflection amount of elastic body **342** increases.

Assuming that the liquid pressure is a variable p and the deflection amount is a function f(p), a relationship of f(p) =a×p+b holds between the liquid pressure and the deflection amount. Where a and b are coefficients. As described above, when the liquid pressure is known, the deflection amount of elastic body **342** is known.

Based on such a relationship between the liquid pressure of viscous liquid **345** and the deflection amount of elastic body **342**, controller **100** of hydraulic excavator **1** calculates the deflection amount of elastic body **342** from the liquid pressure (detection result) of viscous liquid **345** detected by liquid pressure sensor **349**.

Calculation of the inclination angle of hydraulic excavator **1** using the deflection amount of elastic body **342** will be described below.

FIG. **5**A is a view illustrating a state in which hydraulic excavator **1** is positioned on a horizontal plane H. FIG. **5**B is a view illustrating a state in which hydraulic excavator **1** is positioned on an inclined plane SL. In FIG. **5**B, the inclination angle of inclined plane SL with respect to the horizontal plane H is θ. Hereinafter, the case where hydraulic excavator **1** transitions from the state in FIG. **5**A to the state in FIG. **5**B will be described as an example.

In the following description, the case where inclination angle θ of hydraulic excavator **1** is calculated using three liquid-sealed mounts **34***b*, **34***c*, **34***d* among four liquid-sealed mounts **34***a* to **34***d* will be described as an example. Inclination angle θ is an example of an index indicating an "inclination degree" of hydraulic excavator **1**.

FIG. **6** is a view illustrating changes in deflection amounts (displacement amounts) of three liquid-sealed mounts **34***b* to **34***d*. Hereinafter, the description will be given using an xyz-coordinate system (hereinafter, also referred to as "local coordinate system") set in hydraulic excavator **1** together with an XYZ-coordinate system (hereinafter, also referred to as "global coordinate system").

As illustrated in FIG. **6**, points A, B, C, D schematically represent the positions of the upper end surfaces of elastic bodies **342** of four liquid-sealed mounts **34***a* to **34***d* when hydraulic excavator **1** is in the state of FIG. **5**A. In the present example, for convenience of explanation, point D is set as an origin (0,0,0) of the local coordinate system. Hereinafter, the coordinates of points A, B, C in the xyz-coordinate system are (x1, 0, 0), (x1, y1, 0), and (0, y1, 0), respectively. Points B', C', D' schematically represent the positions of the upper end surfaces of elastic bodies **342** of three liquid-sealed mounts **34***b* to **34***d* when hydraulic excavator **1** is in the state of FIG. **5**B.

When hydraulic excavator **1** transitions from the state in FIG. **5**A to the state in FIG. **5**B in a predetermined control cycle Tc (within one control cycle), the position of the upper end surface of elastic body **342** of liquid-sealed mount **34***b* changes from point B to point B'. Similarly, the position of the upper end surface of elastic body **342** of liquid-sealed mount **34***c* changes from point C to point C'. The position of the upper end surface of elastic body **342** of liquid-sealed mount **34***d* changes from point D to point D'.

As described above, the triangle constituted by the three points on the upper end surface of elastic body **342** changes from a triangle BCD to a triangle B'C'D'. That is, the coordinates of each vertex of the triangle (hereinafter, also referred to as "focused triangle") constituted by the three points change. The coordinates of points B', C', D' in the local coordinate system change in the z-axis direction, and for example, become (x1, y1, zb), (0, y1, zc), and (0, 0, zd), respectively.

Controller **100** calculates the inclination angle of triangle B'C'D' with respect to triangle BCD (xy-plane). In this example, triangle BCD is parallel to the XY-plane of the global coordinate system. The inclination angle of triangle BCD with respect to horizontal plane H is 0 degrees. As a result, controller **100** calculates the inclination angle of triangle B'C'D' with respect to horizontal plane H. Specifically, controller **100** calculates inclination angle θ of hydraulic excavator **1** with respect to the horizontal plane based on the direction of the normal vector of triangle B'C'D'.

In this manner, controller **100** calculates the inclination angle of hydraulic excavator **1** by relatively comparing the liquid pressures of three liquid-sealed mounts **34** at the corner of operator's cab **332**. Therefore, the measurement error of inclination angle θ with respect to horizontal plane H of hydraulic excavator **1** is small, and influence of disturbance is hardly received during the calculation of inclination angle θ. For this reason, calibration of the sensor that is conventionally required is also unnecessary.

Specifically, operator's cab **332** is installed on revolving unit **33** with liquid-sealed mount **34** interposed therebetween. Revolving unit **33** is installed on traveling unit **31** with swing circle **32** interposed therebetween. Revolving unit **33** is inclined according to the inclination of traveling unit **31**. Operator's cab **332** is inclined according to the inclination of revolving unit **33**. The liquid pressure of each liquid-sealed mount **34** changes according to the inclination of operator's cab **332**.

Accordingly, as described above, inclination angle θ of hydraulic excavator **1** can be obtained based on the liquid pressure of viscous liquid **345** of each liquid-sealed mount **34**. In particular, even when an attitude of work implement **2**, the load of bucket **23**, the state of the oil in the oil passage (not illustrated), and the state of the additional option of the vehicle body change, operator's cab **332** does not tilt unless revolving unit **33** tilts, so that inclination angle θ can be calculated with high accuracy.

A point P**1** in FIG. **6** indicates a barycentric center position of triangle BCD when hydraulic excavator **1** is in the state of FIG. **5**A. A point P**2** indicates the barycentric center position of triangle B'C'D' when hydraulic excavator **1** is in the state of FIG. **5**B. As hydraulic excavator **1** transitions from the state in FIG. **5**A to the state in FIG. **5**B, the barycentric center position (hereinafter, also referred to as "barycentric center position P of triangle of interest") of the triangle of interest moves from point P**1** to point P**2**. Controller **100** calculates magnitude and direction of the acceleration of the barycentric center movement from point P**1** to point P**2**.

Although details will be described later, controller **100** determines the possibility of overturning of hydraulic excavator **1** using calculated inclination angle θ of hydraulic excavator **1** with respect to horizontal plane H, and the magnitude and direction of the acceleration of the movement of barycentric center position P of the triangle of interest. Barycentric center position P of the triangle of interest is a position in the local coordinate system.

FIG. **7** is a view illustrating the overturning of hydraulic excavator **1**.

As illustrated in FIG. **7**, a straight line **901** indicates an imaginary line (a straight line in the X-axis direction) of an end portion in the Y-axis negative direction in a portion where right crawler belt apparatus **311** is brought into contact with inclined plane SL. Because FIG. **7** is a front view of hydraulic excavator **1**, straight line **901** is represented as a point. A plane **902** is a virtual plane orthogonal to horizontal plane H. Straight line **901** is a line on plane **902**.

Similarly to FIG. **5**B, hydraulic excavator **1** is located on inclined plane SL. The barycentric center position of hydraulic excavator **1** at this time (hereinafter, also referred to as "barycentric center position Q of work machine") is defined as a position Q**1**. In this state, for example, it is assumed that the operator revolves revolving unit **33** in the clockwise direction (Y-axis negative direction in FIG. **7**) and raises work implement **2** and barycentric center position Q of work machine moves from position Q**1** to position Q**2**. Barycentric center position Q of the work machine is a position in the XYZ-coordinate system.

In this case, barycentric center position Q of the work machine moves to a position Q**2** on the opposite side of position Q**1** with respect to plane **902**. For this reason, hydraulic excavator **1** overturns in the direction of an arrow **903** with straight line **901** as the rotation axis.

In order to prevent the generation of the situation, hydraulic excavator **1** previously determines the possibility of the overturning of hydraulic excavator **1** (hazards, signs), and notifies the operator of the possibility of the overturning of hydraulic excavator **1**. Hereinafter, the determination of the possibility of the overturning of hydraulic excavator **1** will be described in more detail.

FIG. **8** is a block diagram illustrating a hardware configuration of hydraulic excavator **1** and a functional configuration of controller **100** of hydraulic excavator **1**.

As illustrated in FIG. **8**, hydraulic excavator **1** includes four liquid-sealed mounts **34***a* to **34***d*, controller **100**, and a notification device **400**.

Liquid-sealed mounts **34***a* to **34***d* include liquid pressure sensor **349**. As described above, liquid pressure sensor **349** detects the liquid pressure of viscous liquid **345** and notifies controller **100** of the detection result.

Notification device **400** includes a lamp **401**, a buzzer **402**, a vibration device **403**, and a monitor device **404**. For example, vibration device **403** is incorporated in an operation apparatus such as an operation lever. Notification device **400** may include at least one of lamp **401**, buzzer **402**, vibration device **403**, and monitor device **404**.

Controller **100** controls the overall operation of hydraulic excavator **1**. Controller **100** is installed in body **3**. Typically, controller **100** is installed in operator's cab **332**.

Controller **100** includes an inclination angle calculation unit **110**, an acceleration calculation unit **120**, a work machine barycentric center calculation unit **130**, and a determination unit **140**. Inclination angle calculation unit **110** includes a deflection amount calculation unit **111**, a vertex coordinate calculation unit **112**, a plane specification unit **113**, and a normal vector calculation unit **114**. Acceleration calculation unit **120** includes a plane barycentric center calculation unit **121** and a movement amount calculation unit **122**.

Inclination angle calculation unit **110** calculates inclination angle θ of hydraulic excavator **1** with respect to horizontal plane H. Inclination angle calculation unit **110** calculates the inclination angle for each control cycle Tc. Hereinafter, processing in inclination angle calculation unit **110** will be described.

Deflection amount calculation unit **111** calculates the deflection amounts of elastic bodies **342** of three liquid-sealed mounts **34** based on detection results of three predetermined liquid pressure sensors **349** among four liquid pressure sensors **349**. In the example of FIG. **6**, deflection amount calculation unit **111** calculates the deflection amount of each elastic body **342** of liquid-sealed mounts **34***b*, **34***c*, **34***d*. Deflection amount calculation unit **111** calculates the deflection amount of each elastic body **342** based on the relationship (FIG. **4**) between the liquid pressure of viscous liquid **345** and the deflection amount of elastic body **342** described above.

Vertex coordinate calculation unit **112** calculates coordinates of three vertices in the local coordinate system of the above-described triangle of interest (see FIG. **6**) based on the deflection amounts of three elastic bodies **342**.

Based on the calculated coordinates of the three vertices, plane specification unit **113** identifies a plane including the three coordinates in the local coordinate system. Normal vector calculation unit **114** calculates a normal vector of the plane specified by plane specification unit **113**.

Inclination angle calculation unit **110** calculates an inclination angle of hydraulic excavator **1** with respect to horizontal plane H based on the calculated normal vector. Inclination angle calculation unit **110** transmits information about the calculated inclination angle to work machine barycentric center calculation unit **130**. Inclination angle calculation unit **110** sends the information about the coordinates of the three vertices calculated by vertex coordinate calculation unit **112** to acceleration calculation unit **120**.

Acceleration calculation unit **120** calculates the magnitude and direction of the acceleration of the barycentric center movement of the above-described triangle of interest (see FIG. **6**). Acceleration calculation unit **120** periodically (every control cycle Tc described above) calculates the magnitude and direction of the acceleration of the barycentric center movement of the triangle of interest.

Plane barycentric center calculation unit **121** periodically calculates barycentric center position P of the triangle of interest based on the coordinates of the three vertices. Movement amount calculation unit **122** periodically calculates the movement amount (change amount) of barycentric center position P of the triangle of interest based on the calculation result of plane barycentric center calculation unit **121**.

Based on the movement amount calculated by movement amount calculation unit **122** and barycentric center position P of the triangle of interest calculated by plane barycentric center calculation unit **121**, acceleration calculation unit **120** periodically calculates the magnitude and direction of the acceleration of the movement of barycentric center position P of the triangle of interest. Acceleration calculation unit **120** transmits the information about the magnitude and direction of the calculated acceleration to work machine barycentric center calculation unit **130**.

Work machine barycentric center calculation unit **130** periodically calculates barycentric center position Q (see FIG. **7**) of the work machine based on inclination angle θ, and the magnitude and direction of the acceleration. Work machine barycentric center calculation unit **130** transmits the information about calculated barycentric center position Q of the work machine to determination unit **140**.

Determination unit **140** determines the possibility of the overturning of hydraulic excavator **1** based on barycentric center position Q of the work machine. Determination unit **140** causes notification device **400** to execute the notification based on the determination result. Specifically, determination unit **140** calculates an index value indicating the possibility of the overturning based on barycentric center position Q of the work machine. Determination unit **140** determines whether the index value exceeds a predetermined threshold. When determining that the index value exceeds the threshold, determination unit **140** transmits a predetermined signal (command) to notification device **400** in order to notify the operator of the warning.

The index value is represented as a difference between calculated barycentric center position Q of the work machine and a predetermined reference barycentric center position (default barycentric center position). As the index value, a component parallel to the XY-plane (horizontal plane H) in the XYZ-coordinate system described above can be mainly used among the differences.

The reference barycentric center position is a barycentric center position when hydraulic excavator **1** is located on horizontal plane H. The reference barycentric center position is a barycentric center position when revolving unit **33** does not revolve. The reference barycentric center position is a barycentric center position when work implement **2** is located at a predetermined position.

When receiving a signal notifying the warning from controller **100**, notification device **400** notifies the operator of the warning. For example, lamp **401** blinks. Buzzer **402** sounds. Vibration device **403** vibrates. Monitor device **404** displays a warning screen on the screen.

The notification device issues the alarm in this manner, so that the operator can know the possibility of the overturning of hydraulic excavator **1**. Therefore, when the operator performs an overturning avoidance operation, hydraulic excavator **1** can be prevented from overturning.

Controller **100** includes a processor (not illustrated) and a memory in which various programs are stored. The processor executes the program to implement inclination angle calculation unit **110**, acceleration calculation unit **120**, work machine barycentric center calculation unit **130**, and determination unit **140**. In addition, all or part of the processing in inclination angle calculation unit **110**, acceleration calculation unit **120**, work machine barycentric center calculation unit **130**, and determination unit **140** may be implemented by a dedicated arithmetic circuit such as an application specific integrated circuit (ASIC).

FIG. **9** is a flowchart illustrating a flow of processing of controller **100**. The following processing is executed for each control cycle Tc.

As illustrated in FIG. **9**, in step S**1**, controller **100** receives the signals that are the detection result from liquid pressure sensors **349** of the plurality of liquid-sealed mounts **34**. In step S**2**, controller **100** calculates inclination angle θ of hydraulic excavator **1** with respect to horizontal plane H based on the signals received from three liquid pressure sensors **349**. In step S**3**, controller **100** calculates the magnitude and direction of the acceleration of hydraulic excavator **1** based on the movement of barycentric center position P of the triangle of interest.

In step S**4**, controller **100** calculates barycentric center position Q of the work machine based on the calculated inclination angle, and the calculated magnitude and direction of the acceleration. In step S**5**, controller **100** calculates the index value indicating the possibility of the overturning based on barycentric center position Q of the work machine and the reference barycentric center position. In step S**6**, controller **100** determines whether the calculated index value exceeds a predetermined threshold.

When determining that the index value exceeds the threshold (YES in step S**6**), controller **100** sends a warning notification instruction to notification device **400** in step S**7**. Thus, notification device **400** notifies the operator of the warning.

In step S**8**, controller **100** restricts the operation of work implement **2**. For example, controller **100** restricts the movement of work implement **2** in the direction in which the possibility of the overturning exists regardless of the operator's operation. Thus, the possibility of the overturning of hydraulic excavator **1** can be reduced. When determining that the index value does not exceed the threshold (NO in step S**6**), controller **100** returns the processing to step S**1**. When next control cycle Tc arrives, controller **100** repeats the processing in and after step S**1**.

A part of the configuration of hydraulic excavator **1** described above is extracted as follows.

Hydraulic excavator **1** includes work implement **2**, traveling unit **31**, revolving unit **33** revolvably supported by traveling unit **31**, and controller **100**. Revolving unit **33** includes frame **331** to which work implement **2** is attached, operator's cab **332**, and a plurality of liquid-sealed mounts **34** that are installed at different positions of frame **331** and support operator's cab **332**. Each liquid-sealed mount **34** includes liquid pressure sensor **349** that detects the liquid pressure of viscous liquid **345** sealed within liquid-sealed mount **34**. Controller **100** calculates inclination angle θ of hydraulic excavator **1** with respect to horizontal plane H based on the detection result of each liquid pressure sensor **349**. According to such the configuration, as described above, inclination angle θ of hydraulic excavator **1** with respect to horizontal plane H can be accurately determined.

Hydraulic excavator **1** further includes notification device **400**. Controller **100** determines the possibility of the overturning of hydraulic excavator **1** based on inclination angle θ. Controller **100** causes notification device **400** to execute the notification based on the determination result. According to such the configuration, the operator of hydraulic excavator **1** can know that the possibility of the overturning of hydraulic excavator **1**.

Controller **100** further calculates the magnitude and direction of the acceleration of hydraulic excavator **1** based on the detection result by each liquid pressure sensor **349**. Controller **100** determines the possibility of the overturning of hydraulic excavator **1** based on inclination angle θ, and the magnitude and direction of the acceleration. According to such the configuration, the possibility of the overturning of hydraulic excavator **1** can be accurately determined as compared with a configuration using only inclination angle θ.

Controller **100** further calculates barycentric center position Q of the work machine based on inclination angle θ, and the magnitude and direction of the acceleration. Controller **100** determines the possibility of the overturning of hydraulic excavator **1** based on barycentric center position Q of the work machine. When the possibility of the overturning is determined based on barycentric center position Q of the work machine of hydraulic excavator **1**, the determination can be performed with high accuracy.

Controller **100** further calculates the index value indicating the possibility of the overturning of hydraulic excavator **1** based on barycentric center position Q of the work machine. When the index value exceeds the threshold, controller **100** causes notification device **400** to issue the warning notification.

When the index value exceeds the threshold, controller **100** restricts the operation of work implement **2**. According to such the configuration, the possibility of the overturning of hydraulic excavator **1** can be further reduced.

Hydraulic excavator **1** includes at least three liquid-sealed mounts **34**. Controller **100** calculates the displacement amount (deflection amount) from the reference value of liquid-sealed mount **34** for each of liquid-sealed mounts **34** based on the liquid pressure of viscous liquid **345** of three liquid-sealed mounts **34**. Controller **100** calculates inclination angle θ based on the three displacement amounts. When the liquid pressure of viscous liquid **345** of three liquid-sealed mounts **34** is used, the above-described triangle of interest can be formed. Therefore, the controller **100** can calculate the inclination angle θ.

<Modifications>

(1) In the above description, controller **100** performs the processing using the detection results of liquid pressure sensors **349** of three liquid-sealed mounts **34***b*, **34***c*, **34***d*. However, the present invention is not limited thereto. Controller **100** may use the detection results of liquid pressure sensors **349** of any three liquid-sealed mounts **34** among four liquid-sealed mounts **34***a* to **34***d*.

(2) Focusing on a plurality of triangles, the inclination angle of each triangle of interest may be obtained, and the average value thereof may be used as inclination angle θ.

(3) Because controller **100** is configured to utilize the detection results of three liquid pressure sensors **349**, one of four liquid-sealed mounts **34** may not include liquid pressure sensor **349**.

(4) In the above description, determination unit **140** of controller **100** compares the index value with one threshold, and determines whether to send the predetermined signal to notification device **400**. However, the present invention is not limited thereto, and determination unit **140** may compare with a plurality of thresholds (for example, Th**2**>Th**1**). In this case, when the index value exceeds threshold Th**1**, determination unit **140** transmits the first signal to notification device **400**. When the index value exceeds threshold Th**2**, determination unit **140** transmits the second signal to notification device **400**.

Notification device **400** makes a notification mode different between when the first signal is received and when the second signal is received. For example, when notification device **400** receives the second signal, the degree of warning is made greater than that when the first signal is received. According to such the configuration, the operator can know that the possibility of the overturning of hydraulic excavator **1** is increased.

(5) In the above description, the work machine is the hydraulic excavator. However, the present invention is not limited thereto, and any work machine including a work implement may be used. For example, the work machine may be a crawler dozer, a loader, or a grader. The work machine includes the body, the work implement, and controller **100**. The body includes the frame to which the work implement is attached, the operator's cab, and the plurality of liquid-sealed mounts **34** that are installed at different positions of the frame and support the operator's cab.

(6) In the above description, it is assumed that the plurality of liquid pressure sensors exist, but one liquid pressure sensor can be used as long as the inclination degree is viewed. A liquid pressure standard value of one liquid pressure sensor is stored, and whether the rubber mount of the pressure sensor is compressed or pulled can be seen from the change from the liquid pressure standard value, and the inclination degree can be seen. Controller **100** can calculate the inclination degree of the work machine based on the change in liquid pressure from the liquid pressure standard value stored previously in the memory of controller **100**.

The above embodiment is only by way of example, and the present invention is not limited to the above embodiment. The scope of the present invention is indicated by the claims, and it is intended that all modifications within the meaning and scope of the claims are included in the present invention.

REFERENCE SIGNS LIST

1: hydraulic excavator, **2**: work implement, **3**: body, **21**: boom, **22**: dipper stick, **23**: bucket, **31**: traveling unit, **32**: swing circle, **33**: revolving unit, **34***a*, **34***b*, **34***c*, **34***d*: liquid-sealed mount, **35**: hydraulic motor, **100**: controller, **110**: inclination angle calculation unit, **120**: acceleration calculation unit, **130**: work machine barycentric center calculation unit, **140**: determination unit, **211**: boom cylinder, **221**: dipper stick cylinder, **231**: bucket cylinder, **242**: boom distal end pin, **243**: dipper stick distal end pin, **311**: crawler belt apparatus, **331**: frame, **331***a*: frame body, **331***b*: beam, **332**: operator's cab, **332***a*: front surface portion, **332***b*: rear surface portion, **332***c*: side surface portion, **332***d*: upper surface portion, **332***e*: floor board bracket, **332***f*: floor **10** board, **341**: casing, **342**: elastic body, **343**: fluid chamber, **344**: resistance plate, **345**: viscous liquid, **346**: compressed air, **347**: shaft, **348**: mounting port, **349**: liquid pressure sensor, **400**: notification device, **901**: straight line, **902**: plane, H: horizontal plane, P: barycentric center position of triangle of interest, Q: barycentric center position of work machine, SL: inclined plane

The invention claimed is:

1. A work machine comprising:
a body;
a work implement; and
a controller,
wherein
the body includes a frame to which the work implement is attached, an operator's cab, and a plurality of liquid-sealed mounts that are installed at different positions of the frame and support the operator's cab,
at least one of the plurality of liquid-sealed mounts includes a sensor that detects a liquid pressure of a sealed liquid, and
the controller calculates an inclination degree of the work machine with respect to a horizontal plane based on a result of the detection by the sensor.

2. The work machine according to claim **1**, wherein the inclination degree is an inclination angle.

3. The work machine according to claim **2**, further comprising a notification device, wherein
the controller determines a possibility of overturning of the work machine based on the inclination angle, and
causes the notification device to execute notification based on a result of the determination.

4. The work machine according to claim **3**, wherein
the controller further calculates magnitude and direction of acceleration of the work machine based on the result of the detection by the sensor, and
determines the possibility of the overturning of the work machine based on the inclination angle, and the magnitude and direction of the acceleration.

5. The work machine according to claim **4**, wherein
the controller further calculates a barycentric center position of the work machine based on the inclination angle, and the magnitude and direction of the acceleration, and
determines the possibility of the overturning of the work machine based on the barycentric center position.

6. The work machine according to claim **5**, wherein
the controller further calculates an index value indicating the possibility of the overturning based on the barycentric center position, and
causes the notification device to execute a warning notification when the index value exceeds a threshold.

7. The work machine according to claim **6**, wherein the controller restricts an operation of the work implement when the index value exceeds the threshold.

8. The work machine according to claim **3**, wherein

the work machine includes at least three of the liquid-sealed mounts, and

the controller calculates a displacement amount from a reference value of the liquid- sealed mount for each of the liquid-sealed mounts based on the liquid pressure of the three of the liquid-sealed mounts, and

calculates the inclination angle based on three of the displacement amounts.

9. The work machine according to claim **1**, wherein

the work machine is a hydraulic excavator,

the body includes a traveling unit and a revolving unit revolvably supported by the traveling unit, and

the revolving unit includes the frame, the operator's cab, and the plurality of liquid-sealed mounts.

10. A method for calculating an inclination degree of a work machine, a body of the work machine including a frame to which a work implement is attached, an operator's cab, and a plurality of liquid-sealed mounts that are installed at different positions of the frame and support the operator's cab,

the method comprising:

detecting a liquid pressure of a liquid sealed in at least one of the plurality of liquid-sealed mounts; and

calculating, by a controller of the work machine, an inclination degree of the work machine with respect to a horizontal plane based on a result of the detection.

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